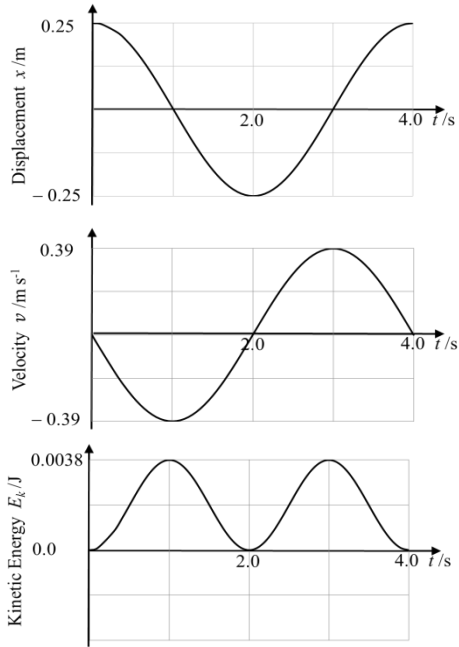
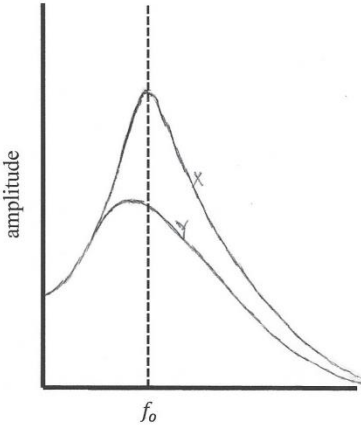


Question			Marking details	Marks available				Maths	Prac
				AO1	AO2	AO3	Total		
7	(a)	(i)	$T = 2\pi\sqrt{\frac{l}{g}}$ $\omega = \frac{2\pi}{T} = \sqrt{\frac{g}{l}}$ combining formulae (1) $= \sqrt{\frac{9.81}{4.0}}$ $= 1.57 \text{ rad s}^{-1}$ (1) convincing		2		2	2	
		(ii)	$v_{\text{max}} = \omega A$ (1) $v_{\text{max}} [= 1.57 \times 0.25] = 0.39 \text{ m s}^{-1}$ (1)		2		2	1	
		(iii)	$E_k = \frac{1}{2}mv^2$ and $v = (-)A\omega\sin\omega t$ / $v = (-)0.39\sin 1.57t$ (1) $E_k = \frac{1}{2} \times 0.05 \times (0.39)^2 \sin^2 1.57t$ (1) $= 3.8 \times 10^{-3} \sin^2 1.57t$ [convincing]	1	1		2	1	
		(iv)	 Phase of the velocity graph (1) Period of the kinetic energy graph (1) Kinetic energy curve always ≥ 0 with shape and phase (1)						

		Values on the axes of both graphs (1)		4		4	2	
		(v) $mgh = 3.8 \times 10^{-3} \text{ J}$ ecf (1) $h = \frac{3.8 \times 10^{-3}}{0.05 \times 9.81} = 0.008 \text{ m}$ (1) (= 8 mm) Or: using trig: $4 - 4 \cos (0.25/4)$ (1) = 8 mm (1)		2		2	2	
(b)	(i)	 amplitude f_0 frequency of driving force Sensible curve [X] with label [accept curve to the origin]	1			1		
	(ii)	Resonance	1			1		
	(iii)	Sensible curve with label (see part (b) (i) [cannot cross X – always below, can co-incide at low frequencies] (1) Lower maximum amplitude at the same frequency or to left of maximum of curve X (1)	2			2		
		Question 7 total	5	11	0	16	8	0